

Face Recognition Under Occlusion

Detailed Project Report with Two Complementary Base Papers

Pgu-Face (Dataset Base) + Chong et al. (Method Base)

1. Executive Summary

This project studies closed-set face identification under real captured facial occlusions. It uses RetinaFace, five-point alignment, a 112×112 crop, pretrained ArcFace/iResNet50, a 512-D embedding and a local 30-way identity head.

Pgu-Face (2016) is used as the dataset/capture base; it introduces 896 images from 224 subjects and does not propose a face-recognition training model. Chong et al. (2022) is the method base; it trains a CNN for feature extraction and uses SVM for label classification.

2. Why Two Base Papers?

Role	Paper	Use
Dataset/capture base	Pgu-Face (2016)	Identity-labelled partially covered data, consumer-camera and multi-session capture reference.
Method base	Chong et al. (2022)	CNN feature extraction + SVM classification for masked-face recognition.
Present work	ArcFace adaptation	Modern pretrained representation, broader occlusions, controlled experiments and live inference.

3. Pgu-Face — Dataset Base

Salari & Rostami (2016), Data in Brief 9, 288–291, DOI 10.1016/j.dib.2016.09.002. The supplied paper is a dataset/Data in Brief article. Its contribution is the Pgu-Face dataset and recording methodology, not a new recognition model.

Pgu-Face contains 896 images from 224 subjects, four images per subject, across up to two sessions separated by at least six days. Conditions are nearly clean-shaven, clean-shaven with sunglasses, stubble/unshaven, and stubble/unshaven with sunglasses. Images were collected with several mobile-phone and commercial cameras, with resolutions ranging from 2 to 26 megapixels.

No recognition CNN, SVM, ArcFace or other proposed training model is reported as the paper's contribution; therefore no model accuracy is attributed to Pgu-Face in this report.

4. Chong et al. — Method Base

Chong, Chong & Ong (2022), ICoICT, DOI 10.1109/ICoICT55009.2022.9914874. The method combines a CNN and SVM: the CNN trains the model and extracts a feature vector, while SVM performs label classification.

RMFD contains 90,000 images/525 categories and LFW-SMFD contains 13,027 images/5,713 categories. The paper uses smaller 50-category subsets for its strongest experiments because SVM does not handle the large category settings well.

Experiment	Labels	Epochs	TAR
001 Testdir	36	100	90.40%
002 RMFD subset	100	100	19.55%
003 RMFD subset	243	100	1.12%
004 LFW	2271	100	9.20%
005 RMFD subset	50	100	98.39%
006 LFW-SMFD subset	50	100	95.24%

The paper reports 98.39% TAR for the 50-category RMFD subset and 95.24% for the 50-category LFW-SMFD subset in Table I/results and conclusion. The abstract instead states 94.29% for LFW-SMFD; this discrepancy is explicitly acknowledged.

The CNN versus CNN+SVM comparison reports 19.56 minutes versus 13.90 minutes of training time, respectively. The paper also reports a 0.01-second testing-time advantage for CNN.

5. Our Project

Our project contains 30 identities, 6,000 reported JPG images and 5,988 successfully processed images. The split is 3,546 train, 1,444 validation and 998 test.

Occlusions include masks, sunglasses, clear glasses, scarves, caps/hats and combinations. The pipeline is RetinaFace → five-point alignment → 112×112 crop → pretrained ArcFace/iResNet50 → 512-D embedding → local identity head.

Pretrained ArcFace is used because the local dataset is small for learning a general face-recognition backbone from scratch. The project focuses local training on identity adaptation.

Experiment	Training	Backbone	Result	Interpretation
E1	Clean-only	Frozen	94.29%	Local baseline
E2	Clean + occluded	Frozen	99.90%	Strongest controlled evidence
E3	Mixed + partial fine-tuning	Last ~15% unfrozen	100% reported	Only +0.10 point over E2; weaker auditability

E1→E2 improves by 5.61 percentage points (99.90 – 94.29). Because the backbone is frozen in both experiments, this is the cleanest evidence that adding occluded training examples improves local adaptation.

6. Detailed Three-Way Comparison

Aspect	Pgu-Face	Chong et al.	Our Project
Paper type	Dataset/Data Article	Recognition method paper	Applied ML system
Primary objective	Create partially covered face dataset	Recognize masked faces using CNN + SVM	Identify known people under multiple real occlusions
Identities	224	RMFD 525 / LFW-SMFD 5713 full; main experiments 50	30
Images	896	90,000 RMFD; 13,027 LFW-SMFD; subsets used for experiments	6,000 reported / 5,988 processed
Occlusion conditions	Sunglasses + facial-hair conditions	Primarily masks	Masks, sunglasses, clear glasses, scarves, caps/hats, combinations
Capture source	Several consumer/mobile and commercial cameras	Public benchmark datasets	Local real capture
Face detection	Not a proposed pipeline	Not central to method	RetinaFace
Alignment	Not proposed as recognition pipeline	Not central	Five-point landmark alignment
Feature representation	No proposed recognition representation	CNN feature vector	512-D ArcFace embedding
Recognition model	None proposed	CNN + SVM	Pretrained ArcFace + 30-way local head
Training model	No recognition training model	CNN trained for feature extraction; SVM classifies labels	Head training; optional partial backbone fine-tuning
Training data strategy	Dataset construction	Masked + unmasked customized training	Clean-only vs mixed-condition vs partial fine-tuning
Epochs	Not applicable	100 in reported experiments	Early stopping for E1/E2; E3 fine-tuning
Best reported recognition result	Not applicable	98.39% TAR RMFD 50-category; 95.24% TAR LFW-SMFD 50-category in results/conclusion	99.90% E2; 100% E3 reported
Scalability	Dataset-oriented	Strong degradation with more SVM categories	Not tested as large-scale/open-set
Deployment	None	Research evaluation	Live phone-browser + Flask HTTPS
Main value to our project	Dataset/capture precedent	Algorithmic/method precedent	Modern adaptation + broader occlusion + deployment

7. Contributions Relative to Both Bases

- 1) Dataset extension: approximately 6,000 local images versus 896 in Pgu-Face, while serving a smaller 30-identity closed-set scope.
- 2) Broader real occlusions: masks, sunglasses, clear glasses, scarves, caps/hats and combinations.
- 3) Modern representation: pretrained ArcFace/iResNet50 rather than the CNN+SVM pipeline.
- 4) Explicit RetinaFace detection and five-point alignment.
- 5) Controlled E1/E2 experiment isolating the effect of occluded training data.
- 6) Measured +5.61 percentage-point gain from E1 to E2.
- 7) E3 partial backbone fine-tuning experiment.
- 8) Live Flask HTTPS phone-browser inference.

8. Research Gap and Positioning

Pgu-Face provides a real-data/capture precedent but no recognition algorithm. Chong et al. provides a recognition algorithm but shows strong dependence on smaller category counts because of SVM scalability.

Our project adapts a strong pretrained face representation to a small local population, broadens the real occlusion conditions, measures clean-versus-occluded training, and demonstrates live inference.

The safest claim is practical adaptation and experimental evaluation, not a new recognition architecture or direct superiority over published benchmark results.

9. Limitations

Small 30-identity population; closed-set task; local camera/lighting distribution; possible split-overlap concern requiring audit; weaker artifact support for E3; no formal open-set stranger rejection; no anti-spoofing; no guarantee for unseen occlusions.

Published paper percentages are not directly comparable with the project's accuracy because datasets, identity counts and evaluation protocols differ.

10. Presentation Narrative

“Pgu-Face is our dataset/capture reference; Chong et al. is our method reference. We modernize the recognition stage with pretrained ArcFace, use RetinaFace alignment, cover more real occlusion types, compare clean-only and mixed-condition adaptation, and deploy the model through a live phone-to-laptop application.”

“Our strongest controlled result is 94.29% to 99.90%, a 5.61 percentage-point improvement. We treat the 100% E3 result as encouraging but not the main evidence because its auditability is weaker and its gain over E2 is only 0.10 points.”

11. Viva Questions

Q: Does Pgu-Face train a recognition model? A: No. It is a dataset/Data in Brief paper.

Q: What does Chong et al. train? A: A CNN for feature extraction; SVM performs label classification. Experiments use 100 epochs.

Q: Why did Chong et al. use 50-category subsets? A: Their results show SVM performance drops strongly with more categories.

Q: What is our strongest result? A: E2 = 99.90%, versus E1 = 94.29%, a +5.61 percentage-point improvement.

Q: Can we claim 99.90% beats Chong's 98.39%? A: No; the datasets and protocols differ.

Q: Is the project open-set? A: No; it is closed-set identification among 30 enrolled identities.

Q: What is the contribution? A: Practical ArcFace adaptation, broader real occlusions, controlled local evaluation and live deployment.

12. References

- Salari, S. R., & Rostami, H. (2016). Pgu-Face: A Dataset of Partially Covered Facial Images. *Data in Brief*, 9, 288–291. DOI: 10.1016/j.dib.2016.09.002.
- Chong, L. W. J., Chong, S.-C., & Ong, T.-S. (2022). Masked Face recognition Using Support Vector Machine and Convolutional Neural Network. 2022 10th International Conference on Information and Communication Technology (ICoICT), 270–274. DOI: 10.1109/ICoICT55009.2022.9914874.
- Deng, J., Guo, J., Xue, N., & Zafeiriou, S. (2019). ArcFace: Additive Angular Margin Loss for Deep Face Recognition. *CVPR*.
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